

Precision Genome Editing with CRISPR-Cas9: Therapeutic Applications and Off-Target Mitigation

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ABSTRACT

The CRISPR-Cas9 bacterial endonuclease system has transformed targeted genomic engineering in biomedical research. We characterize high-fidelity Cas9 variants engineered to eliminate non-specific off-target double-strand breaks. In vivo preclinical trials demonstrate efficient correction of monogenic disease mutations in human hematopoietic stem cells without detectable chromosomal translocations or oncogenic mutations.

Keywords: CRISPR-Cas9, Gene Editing, Genomics, Molecular Therapeutics, Genetic Diseases

1. Introduction

Genetic disorders caused by point mutations historically lacked curative treatments. The CRISPR-Cas9 system utilizes a synthetic single guide RNA (sgRNA) to direct Cas9 endonuclease to complementary genomic loci.

2. Engineering High-Fidelity Cas9

Through rational structure-guided mutagenesis of the DNA-binding pocket, we engineered variants that require stringent RNA-DNA pairing, effectively eliminating off-target cleavage at non-homologous sites.

3. Preclinical In Vivo Efficacy

Targeting the beta-globin locus in patient-derived CD34+ stem cells yielded over 80% on-target editing efficiency. Deep whole-genome sequencing confirmed zero detected off-target alterations above baseline background noise.

4. Conclusion

High-precision CRISPR-Cas9 platforms establish a reliable therapeutic avenue for permanent genetic correction of debilitating hereditary hematologic and immunologic diseases.